

SIMATS ENGINEERING

ASSESSMENT TOOL 3: Mock Interview

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED

CO5: *Design big data processing systems using the Hadoop ecosystem including HDFS, MapReduce, and YARN.*
(BL5)

Dave's Taxonomy: Articulation (Level 4)
Question Count: 10

Total Marks: 25

Code of Conduct

I G. Manobhi Ram (192311322) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: G. Manobhi Ram

Assessment Tasks

Mock Interview

Answer the following interview-oriented questions clearly and concisely. Support your responses with architecture-level reasoning wherever appropriate.

1. Explain the difference between HDFS and traditional file systems.
2. Why is replication important in distributed storage systems?
3. How does MapReduce divide work between map and reduce phases?
4. What role does YARN play in large-scale Hadoop processing?
5. Differentiate NAS and SAN in an enterprise data environment.
6. What is the purpose of ETL in a big data pipeline?
7. How does Apache NiFi help in data integration workflows?
8. What are the key failure points in a Hadoop cluster and how would you handle them?
9. How would you design a secure data ingestion pipeline for a large organization?
10. How would you explain a scalable Hadoop-based processing system to a non-technical manager?

Mock Interview Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Technical Knowledge	25%	Shows deep and accurate technical understanding	Good technical understanding	Basic understanding	Weak understanding
Problem Solving	20%	Responds with strong architecture-level reasoning	Reasonable reasoning	Basic reasoning	Weak reasoning
Communication	20%	Answers are confident, precise, and professional	Mostly clear communication	Moderate clarity	Weak communication
Practical Awareness	20%	Connects concepts to real-world implementation well	Some practical relevance	Limited practical relevance	No practical linkage
Confidence and Professionalism	15%	Highly confident and professional	Generally professional	Moderately professional	Low confidence and professionalism

1. Explain the difference between HDFS and traditional file systems.

Answer:

HDFS is designed to store very large amounts of data across multiple computers. Traditional file systems usually store data on a single computer or storage system. HDFS provides distributed storage, replication, and fault tolerance.

2. Why is replication important in distributed storage systems?

Answer:

Replication means keeping multiple copies of the same data on different machines. If one machine fails, another copy can be used. Therefore, replication provides fault tolerance, reliability, and high availability.

3. How does MapReduce divide work between map and reduce phases?

Answer:

The Map phase processes the input data and produces key-value pairs. The Reduce phase collects the same keys and combines or processes their values to produce the final result. This allows large tasks to be processed in parallel.

4. What role does YARN play in large-scale Hadoop processing?

Answer:

YARN stands for **Yet Another Resource Negotiator**. It manages the resources of a Hadoop cluster and schedules jobs. It allocates CPU and memory to different applications so that multiple jobs can run efficiently.

5. Differentiate NAS and SAN in an enterprise data environment.

Answer:

NAS provides **file-level storage** through a network and is accessed using files and folders. SAN provides **block-level storage** and is mainly used for high-performance applications. Simply, NAS is like a network file server, while SAN works like a remote disk.

6. What is the purpose of ETL in a big data pipeline?

Answer:

ETL stands for **Extract, Transform, and Load**. It extracts data from different sources, cleans or transforms the data, and loads it into a database or storage system. It helps prepare data for analysis.

7. How does Apache NiFi help in data integration workflows?

Answer:

Apache NiFi is a data integration and data-flow tool. It collects data from different sources, processes or transforms it, and sends it to different destinations. It also provides monitoring and management of data flows.

8. What are the key failure points in a Hadoop cluster and how would you handle them?

Answer:

Common failure points include **DataNode failure, NameNode failure, disk failure, and network failure**. Hadoop handles these using data replication, fault tolerance, NameNode high availability, and automatic recovery. Monitoring can also be used to detect failures.

9. How would you design a secure data ingestion pipeline for a large organization?

Answer:

I would use **authentication, authorization, encryption, and data validation**. Data should be encrypted during transmission and storage. I would also use proper access control, logging, and monitoring to protect the data and detect security problems.

10. How would you explain a scalable Hadoop-based processing system to a non-technical manager?

Answer:

I would explain Hadoop as a **team of computers working together**. A large amount of data is divided into

smaller parts and processed by many computers at the same time. When the data increases, we can add more computers, making the system scalable and reliable.